

Exynos 9 Octa 8895

The Exynos 9 Octa 8895 is fabricated using a 10nm process, following in the footsteps of its competitors. It is 27% faster than its predecessor and is almost 40% more power efficient. The 8 cores are divided into 2 clusters, one consisting of 4 custom CPU cores responsible for handling the more intensive tasks, and the remaining 4 Cortex-A53 for less resource intensive tasks.

There are a number of other features packed in the chip including a Mali-G71 GPU that is capable of recording and playing videos in 4K UHD at upto 120fps (well these are some mind boggling figures). It comes with a DPU for features such as the iris and fingerprint scanner.

Primary CPU	4 x 2.5GHz Exynos M2 + 4 x 1.7GHz Cortex-A53
CPU architecture	Custom
CPU structure	RISC
DSP type	VPU
GPU	ARM Mali-G71 MP20
Year Released	2017
Devices	Samsung Galaxy S8

Exynos 8 Octa 8890

It was the first SoC by Samsung to feature its own custom CPU core. Built using a 14nm process. 4 out of the 8 cores are custom designed by Samsung based on the 64 bit ARM v8 structure and are used for resource intensive tasks, while the remaining 4 cores are standard Cortex-A53 cores, that are more power efficient but less powerful making them suitable for various common tasks. Coupled with one of the latest Mali GPUs, the device also supports 4k UHD video recording and playback.

Primary CPU	4x A53@1.586GHz + 4x Exynos M1 @ 2.60GHz (1-2 core load) , 2.29GHz (3-4 core load)
CPU architecture	big.LITTLE
CPU structure	RISC
DSP type	ARM NEON
GPU	ARM Mali-T880
Year Released	2016
Devices	Samsung Galaxy S7, galaxy S7 Edge